RESPIRATORY VIRUS SURVEILLANCE | DATABASE DESIGN REPORT

Respiratory Virus Surveillance Database

A reproducible SQLite model for outbreak monitoring, data quality, and SQL retrieval

SYNTHETIC DATA ONLY - No real people, clinics, surveillance observations, or downloaded datasets are used.

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Git repository: https://github.com/vinayyasa28/data.mining

Purpose and scope

This study draft models a fictional respiratory-virus monitoring programme across four invented regions and eight collection sites. Its 26-week window runs from 1 October 2025 to 31 March 2026. The linked tables support weekly positivity summaries, outbreak alert examples, and practice joining laboratory results to specimen and site context.

Every observation is generated locally from explicit assumptions with Python's seeded random-number generator. No external data are downloaded. The seasonal curves, test mix, and subgroup differences make the records varied enough for database exercises, but they are not calibrated estimates and must not be interpreted as real prevalence or clinical evidence.

Generation at a glance

2,400	10,177	1,500	860
specimens	laboratory results	fictional participants	site-virus-week summaries

The fixed default seed is 20261106. Site size affects sampling weight; each specimen receives a four-virus PCR panel, with occasional repeat PCR and antigen confirmation. Positivity varies by pathogen, week, and site setting. The generator also creates broad age bands, risk groups, vaccination histories, symptom measures, and turnaround times. Rebuilds with the same seed and Python version reproduce the same records.

Measurement scales represented

Scale	Example fields	Why it fits
Nominal	site type; pathogen; result category	Named categories have no inherent order.
Ordinal	age band; symptom severity; alert level	Categories have a meaningful order, but unequal steps.
Interval	temperature_celsius	Celsius differences are meaningful; zero is arbitrary.
Ratio	viral load; turnaround hours; counts	A meaningful zero allows ratios as well as differences.

What makes the records varied

Collection and season

Each specimen date is sampled with a modest winter-volume rise. Virus-specific seasonal curves and site-setting factors shape detection probabilities. These choices create a plausible-looking teaching dataset, not epidemiological estimates.

Tests and measures

Every specimen receives a four-pathogen PCR panel. A small share has repeat PCR, and some positive screens receive antigen confirmation. Simulated Ct moves inversely with viral load; turnaround differs by method.

Generation workflow

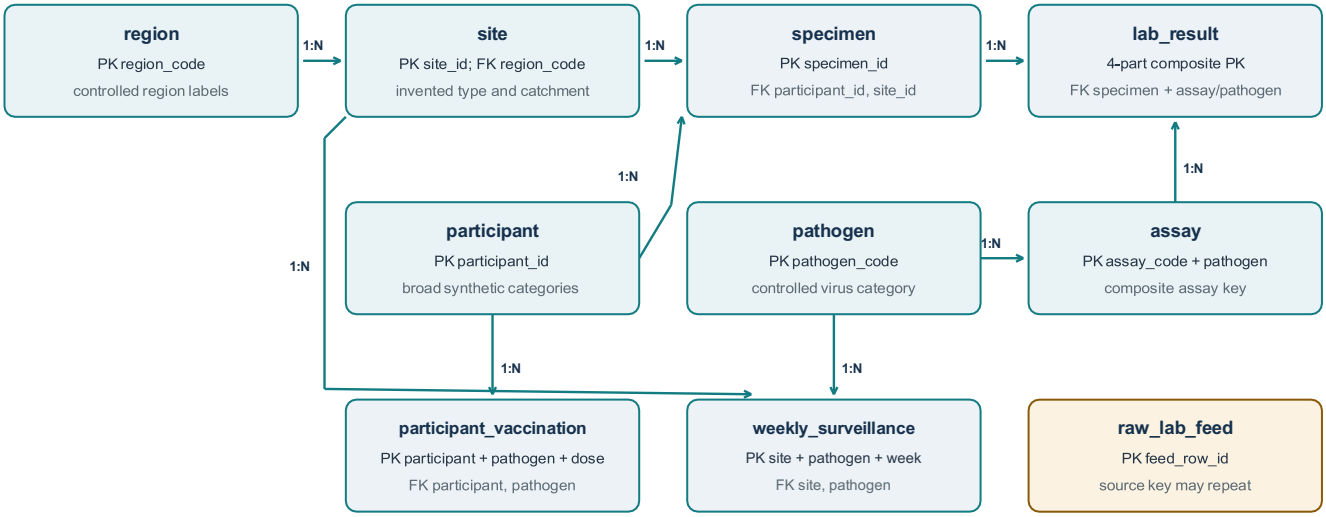
01 Seed	02 Sample	03 Test	04 Preserve and summarize
Initialize Python's random generator with a fixed seed for repeatable output.	Create 1,500 fictional profiles and select 2,400 dated specimens across sites.	Generate four-virus PCR panels, seasonal detection patterns, repeat tests, and selected antigen results.	Add documented NULLs and repeated feed keys, then calculate site-pathogen-week summaries.

02 / RELATIONAL DESIGN

Schema and design choices

Entity relationship diagram

Boxes list primary and foreign keys. Arrows show one-to-many links; vaccination history and weekly summaries use composite primary keys.



Arrows show 1:N links. Bridge boxes list both parent FKs; raw_lab_feed is separate staging without enforced relationships.

Table and attribute choices

Table	Attributes and constraint rationale
region	PK region_code; unique region_name; controlled climate band keeps grouping labels consistent.
site	PK site_id; FK region_code; unique invented name, category site_type, and positive catchment population.
participant	PK participant_id; ordinal age and risk bands plus enrolment group; synthetic integer IDs only.
pathogen	PK pathogen_code; unique name, family, and bounded ordinal severity value.
assay	Composite PK (assay_code, pathogen_code); FK pathogen; constrains valid method-virus pairs.
specimen	PK specimen_id; FKs participant/site; valid dates, onset before collection, severity consistency, Celsius range.
lab_result	Four-part PK (specimen, method, pathogen, replicate); FK specimen and composite assay FK; bounded Ct/time and nonnegative load.
participant_ vaccination	Composite PK (participant, pathogen, dose); both FKs, bounded dose number, valid date, controlled product.
weekly_surveillance	Composite PK (site, pathogen, week); FKs site/pathogen; count, rounded-rate, and alert thresholds checked.
raw_lab_feed	PK feed_row_id; source_record_key deliberately non-unique; nullable receipt time preserves intake defects.

Three composite primary keys prevent duplicate laboratory events, vaccination doses, and site-pathogen-week summaries. The lab_result composite foreign key enforces an available assay/pathogen pair. `raw_lab_feed` retains repeated source keys; curated tables enforce their own keys.

03 / QUALITY AND RESPONSIBLE USE

Data quality, ethics, and student review

The curated analytical tables enforce keys and relationships. A separate raw staging table preserves realistic source imperfections for inspection.

Deliberate data-quality conditions

Condition	Default-seed count	Interpretation
Symptomatic onset date missing	322	Recording gap; asymptomatic onset dates are not applicable (378 rows).
Symptomatic temperature missing	123	Intake gap; asymptomatic temperatures are not applicable (344 rows).
Detected PCR missing Ct / viral load	41 / 56	Missing among positive PCR results; negatives have no applicable numeric measurement.
Raw source key repeats	250 groups / 250 extra rows	Exact repeated intake events remain in raw_lab_feed for deduplication practice.
Raw receipt timestamp missing	351	Source-feed timing is incomplete; ingestion time is retained separately.

For quality work, first profile missingness by symptom status so structural non-applicability is not confused with a recording error. The raw source key is deliberately non-unique, while a surrogate feed-row key preserves each arrival. Curated lab results use their composite key to prevent repeated specimen-assay-pathogen-replicate events. The generator and notebook inspect row counts, duplicate groups, missing fields, panel completeness, weekly summaries, foreign keys, and SQLite integrity.

Ethics and privacy

The database uses generated integers, broad age bands, invented site names, and no real identifiers, addresses, or downloaded observations. Health-shaped data can still encourage misleading conclusions: all rates and subgroup patterns are consequences of chosen simulation assumptions. A real surveillance system would require a lawful basis, appropriate consent or other authority, data minimization, access controls, retention limits, transparent purpose, and a disclosure-risk review before sharing. Re-identification risk must be assessed even after direct identifiers are removed.